

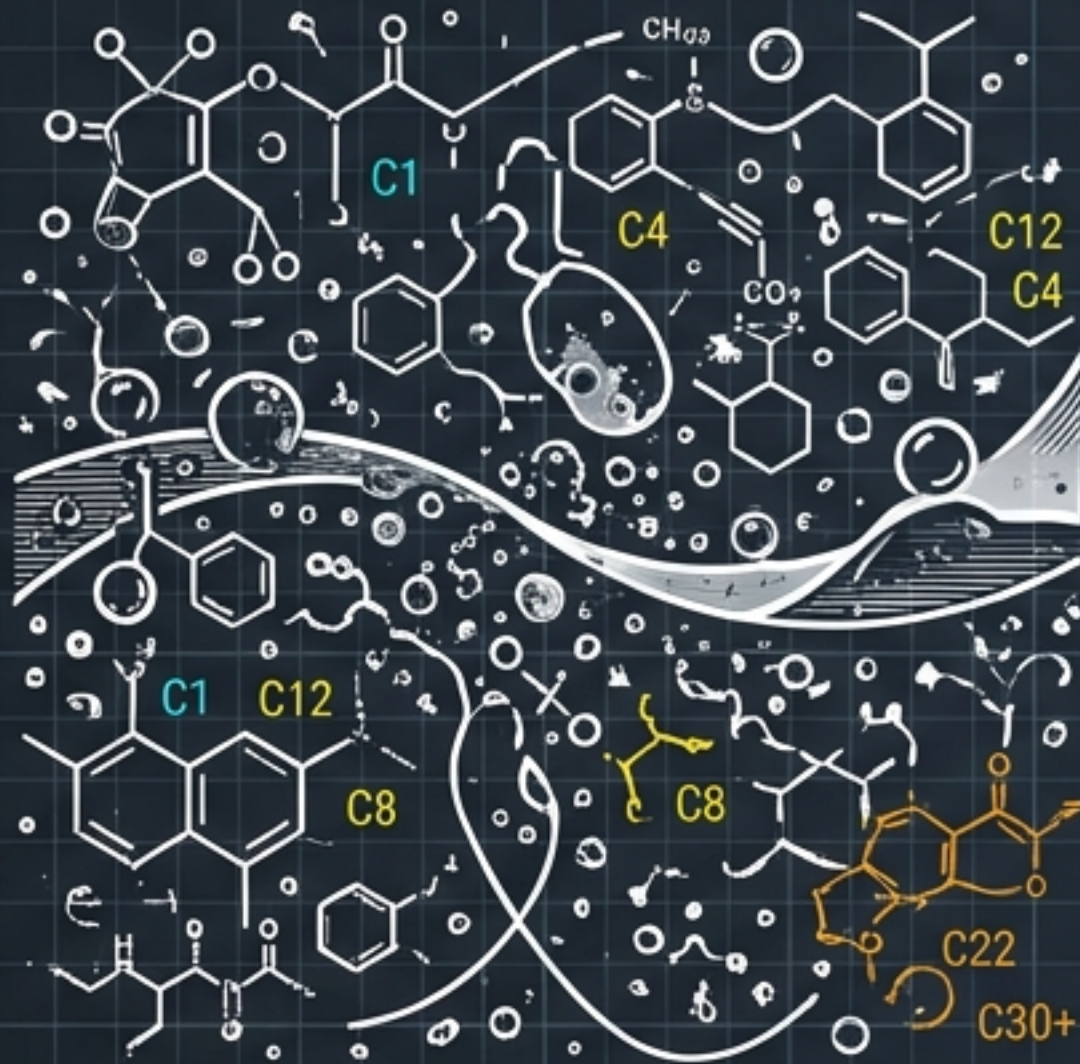
Ternary Phase Diagrams in Petroleum Engineering

Mapping Phase Behavior, Binodal
Curves, and Critical Points

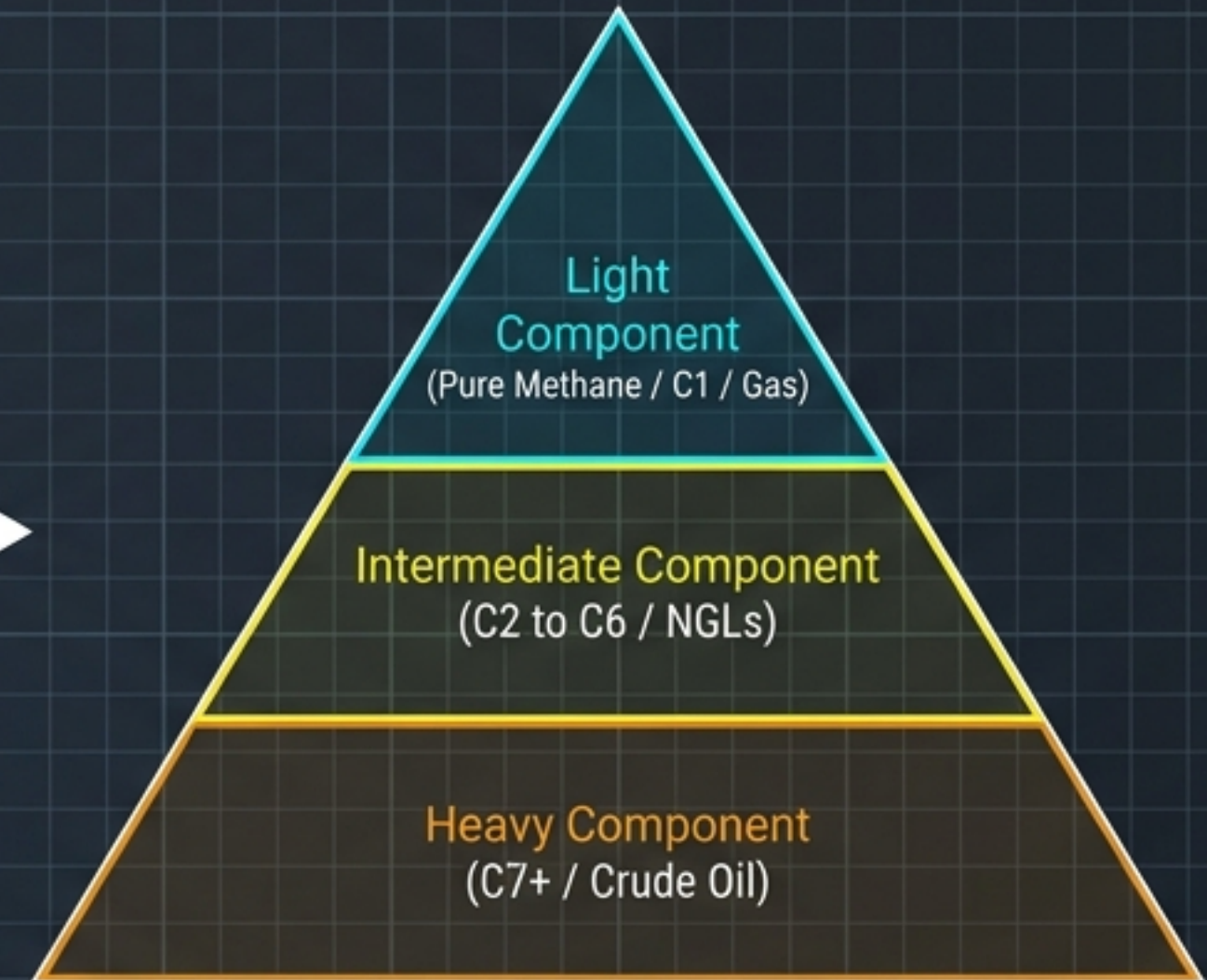


Condensing thousands of hydrocarbon molecules into three manageable pseudo-components

Reservoir Fluid Complexity



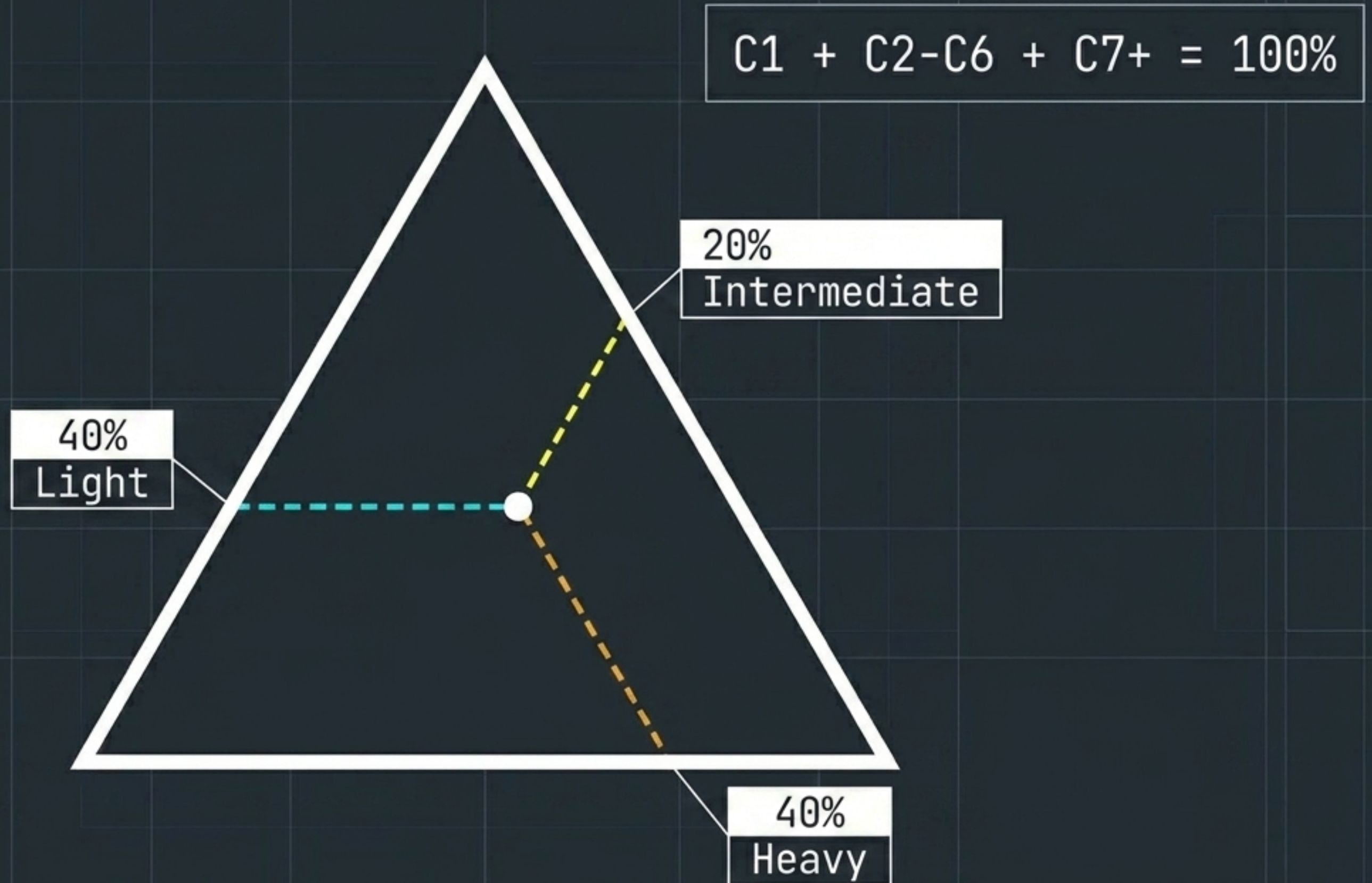
Mathematical Stage



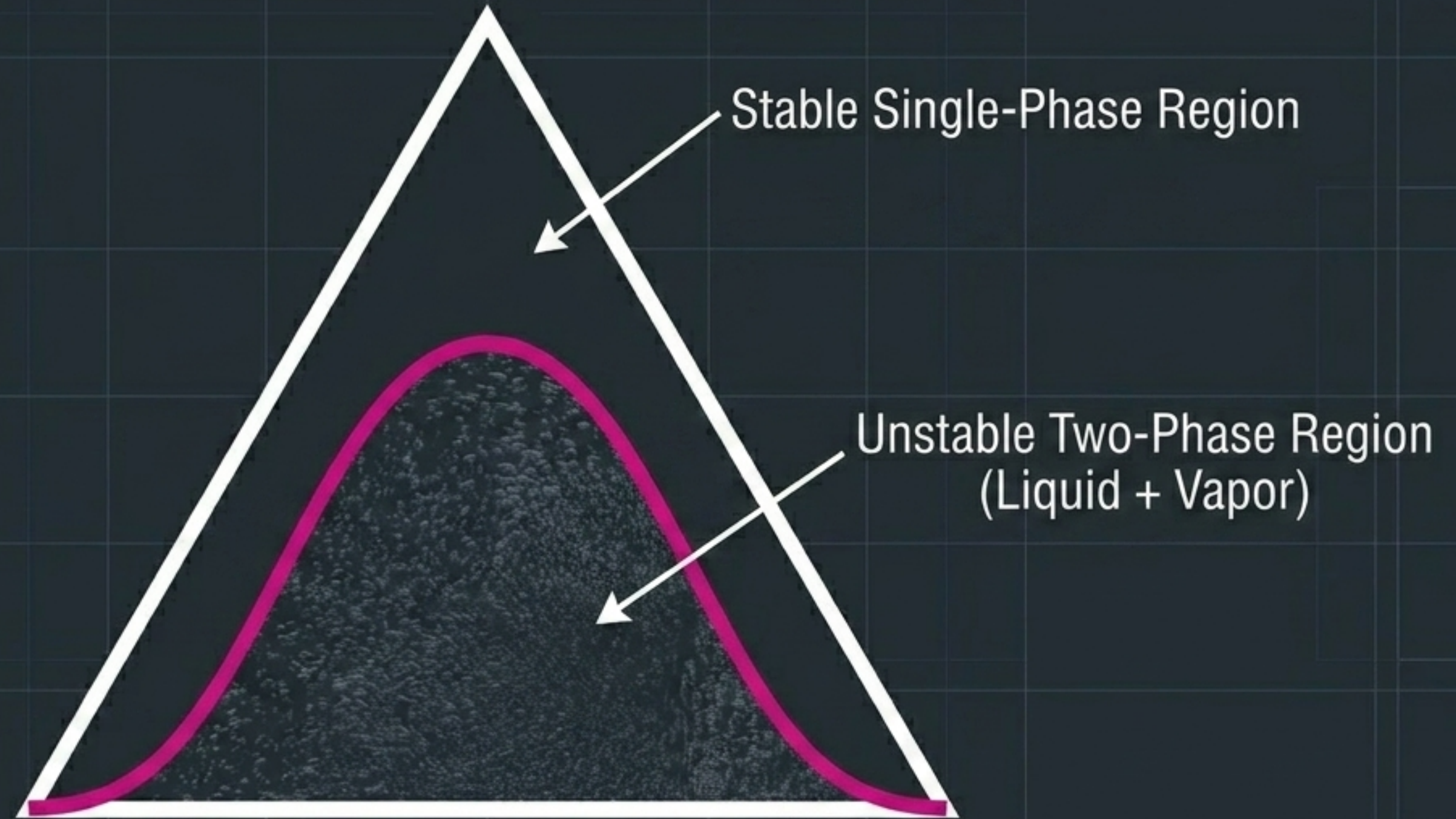
The equilateral triangle defines the absolute concentration boundaries of the fluid system



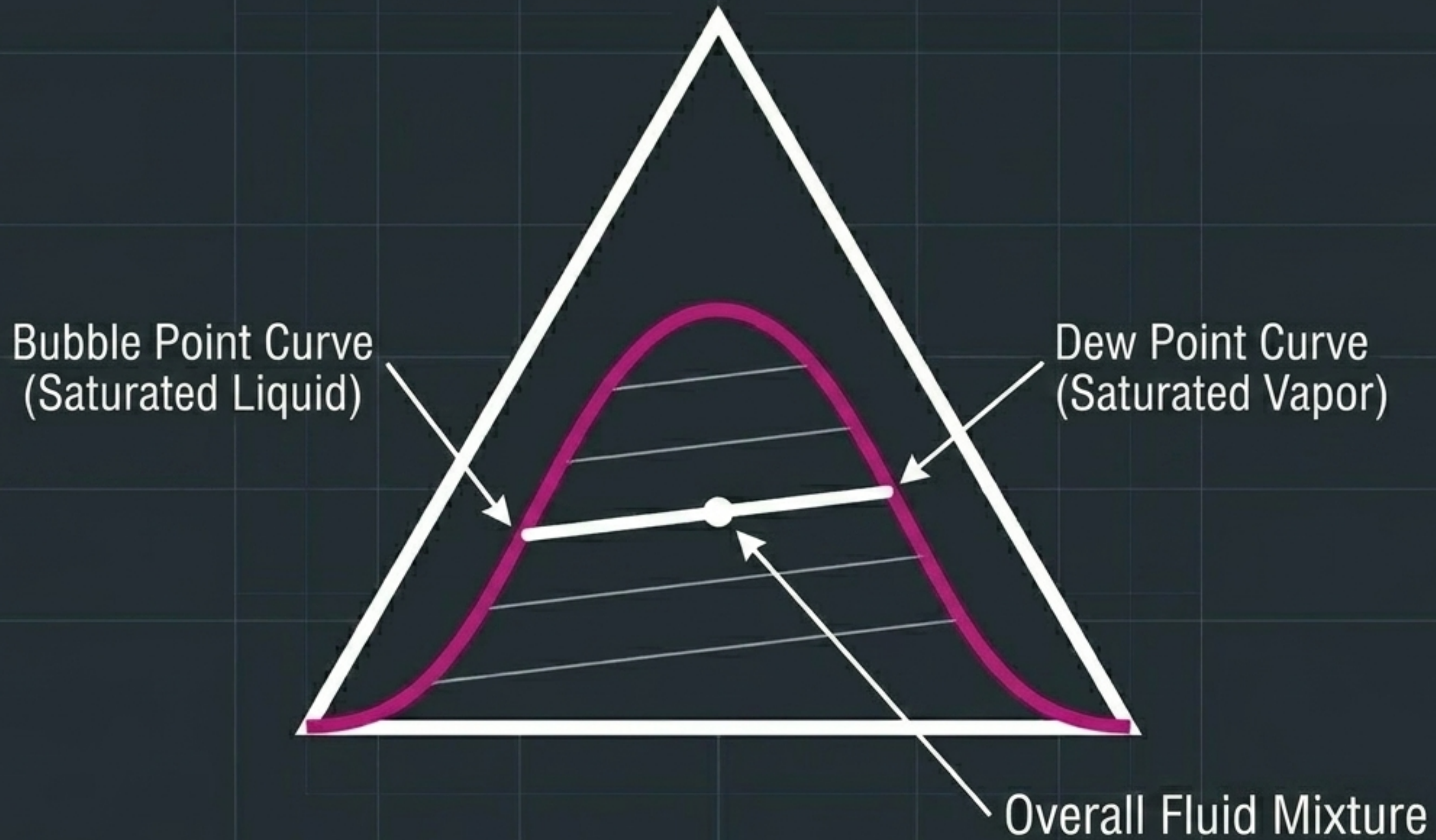
Mixture coordinates are plotted parallel to the opposite zero-concentration axis



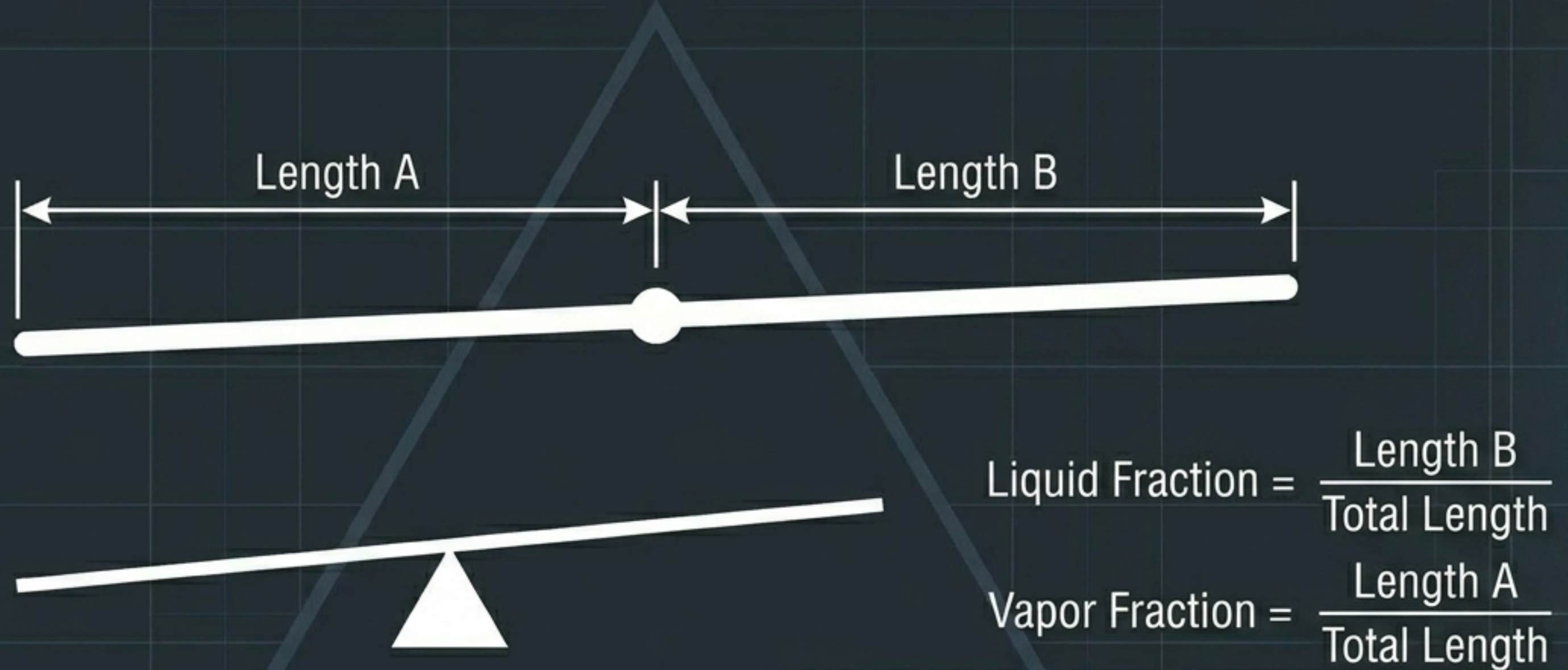
The binodal curve divides the stable single-phase fluid from the separating two-phase region



Tie lines connect the equilibrium compositions of separated liquid and gas phases

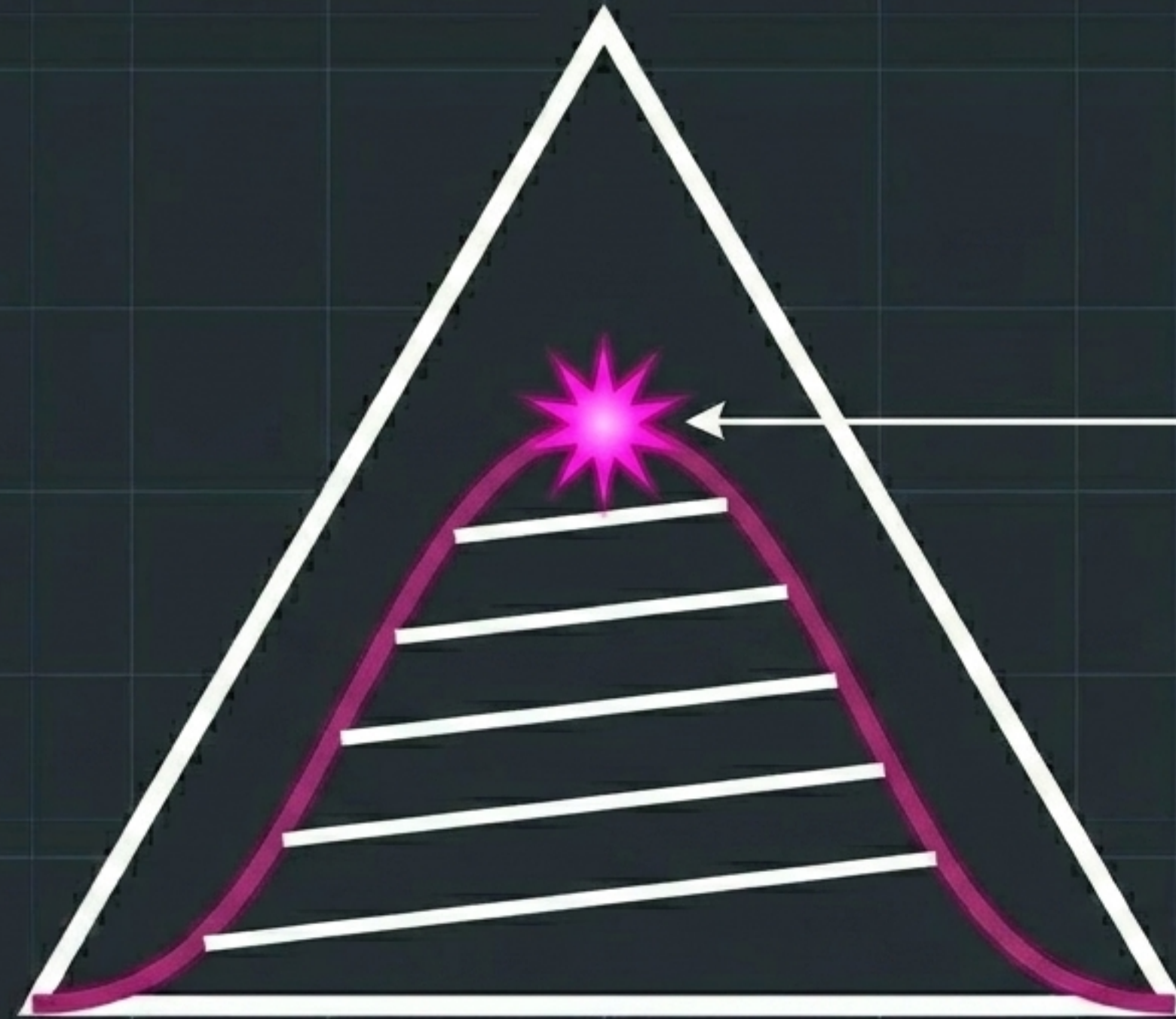


The Inverse Lever Rule calculates the exact volumetric fractions of liquid and vapor



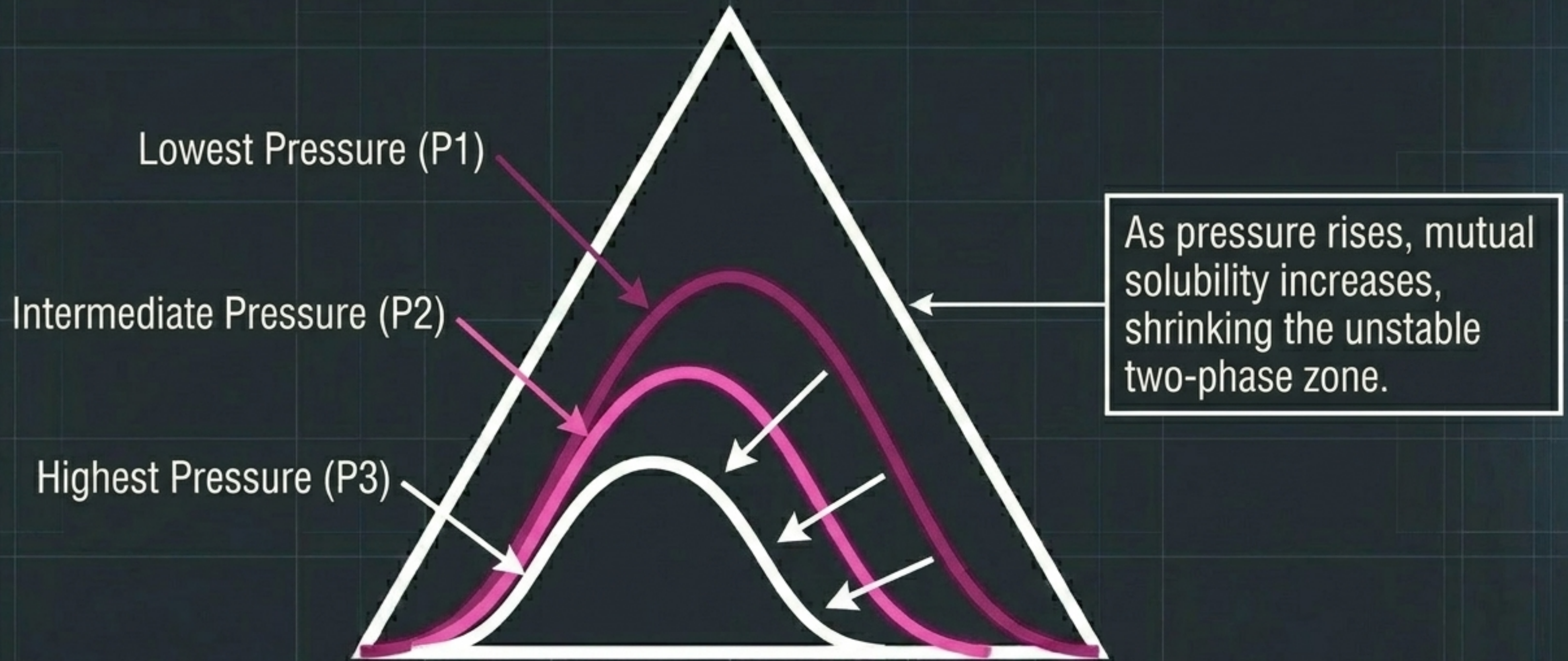
Counter-intuitive principle: The closer the mixture point is to a phase boundary, the higher the fraction of that phase.

The plait point marks the exact critical boundary where liquid and gas become indistinguishable



1. Point Definition:
Tie line length shrinks to absolute zero.
2. Thermodynamic State:
Intensive properties (density, viscosity) of liquid and vapor phases become identical.
3. Phase Distinction:
Ceases to exist entirely.

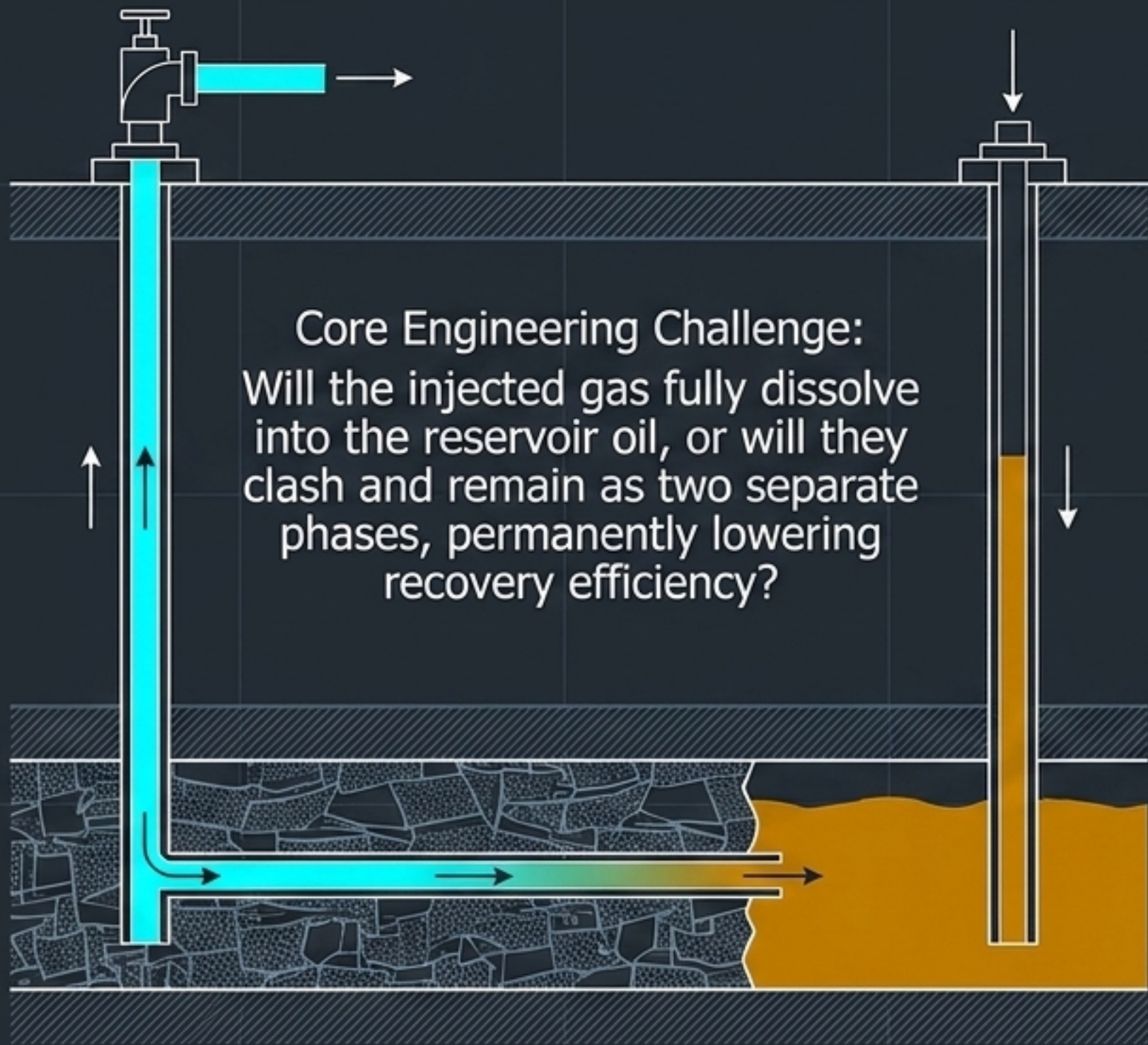
Increasing reservoir pressure forces the two-phase envelope to collapse



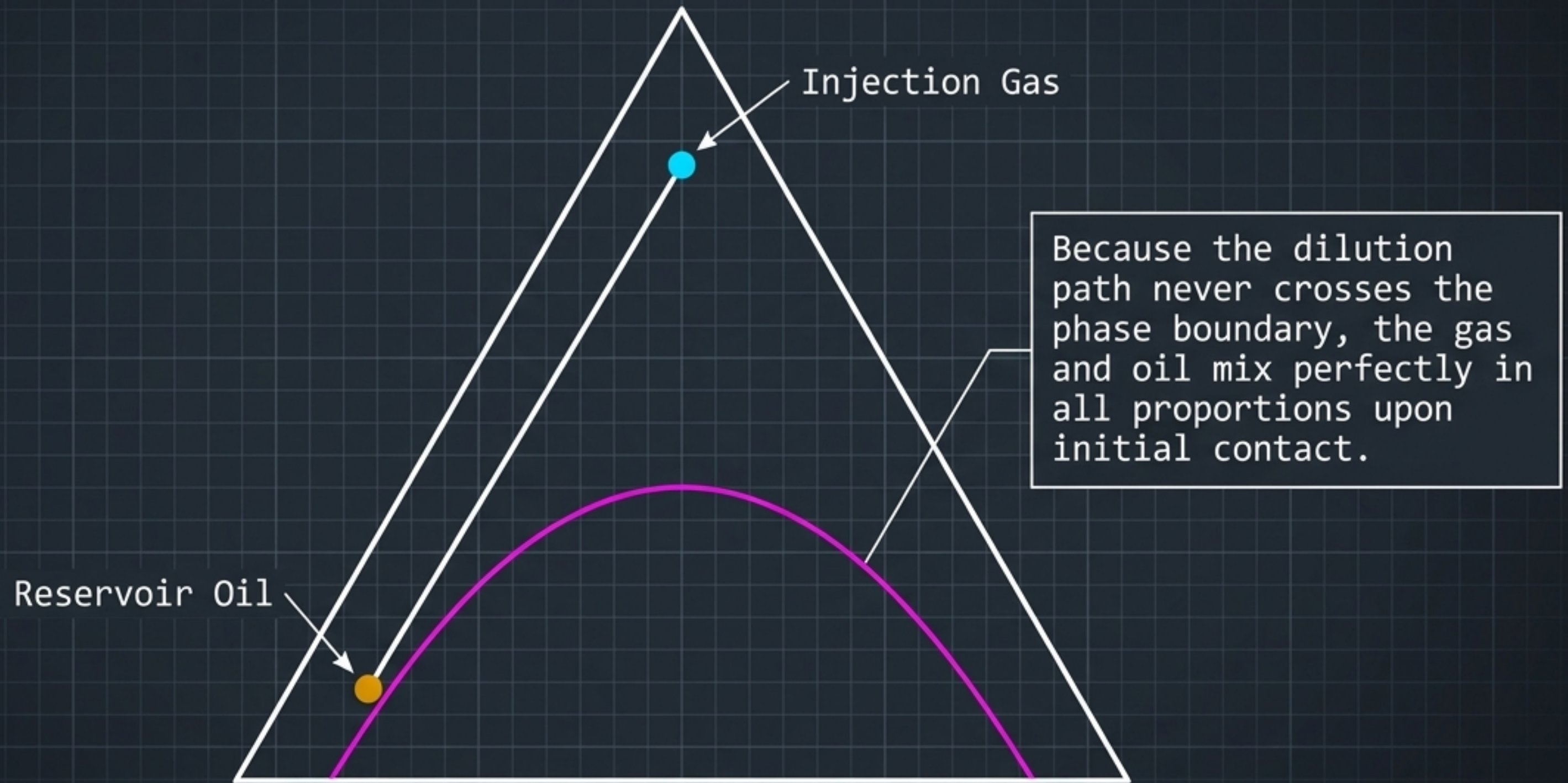
Decoding the thermodynamic blueprint: Geometric features vs. Physical meaning

Visual Indicator	Geometric Term	Physical Reality
	Binodal Curve	Phase Boundary (separates miscible from immiscible states)
	Tie Line	Equilibrium Split (connects liquid and vapor compositions)
	Plait Point	Critical State (zero distinction between phases)

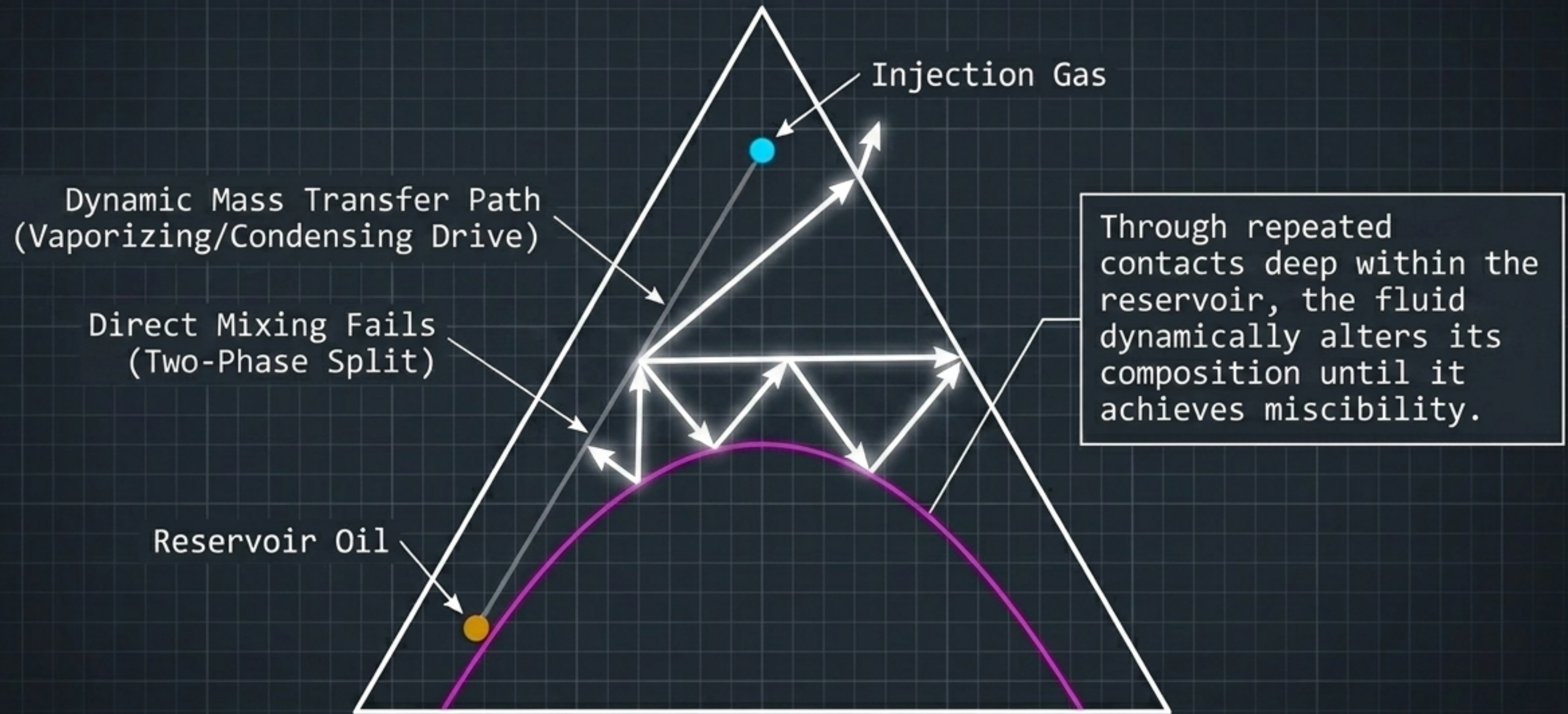
Ternary diagrams engineer the precise design of miscible gas injection strategies



First-contact miscibility achieves instant single-phase flow by bypassing the two-phase envelope

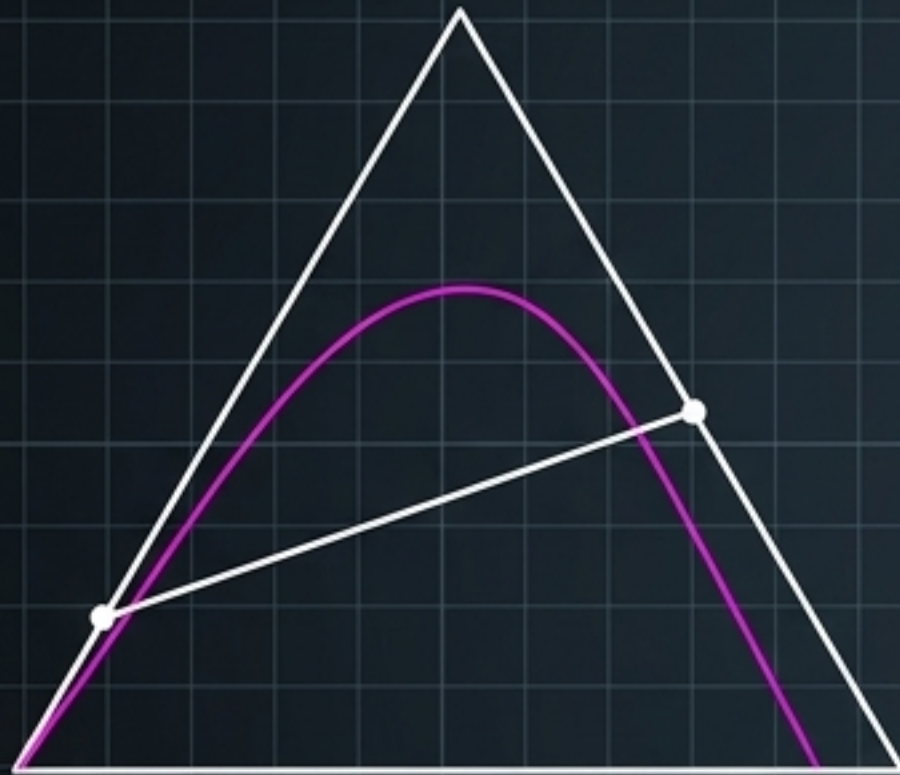


Multi-contact miscibility utilizes dynamic mass transfer to step over the phase boundary



Contrasting gas injection strategies across the thermodynamic blueprint

Immiscible Drive



Text Verdict

Fluids remain physically separate. Low recovery.

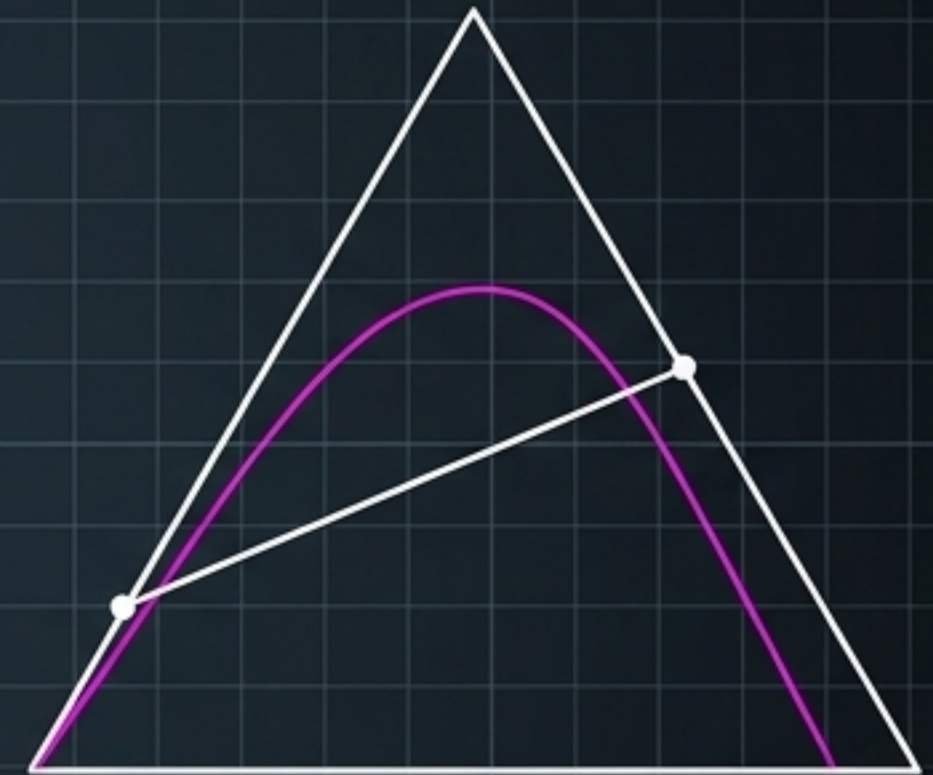
Multi-Contact Miscible (MCM)



Text Verdict

Miscibility achieved dynamically over time and distance. High recovery.

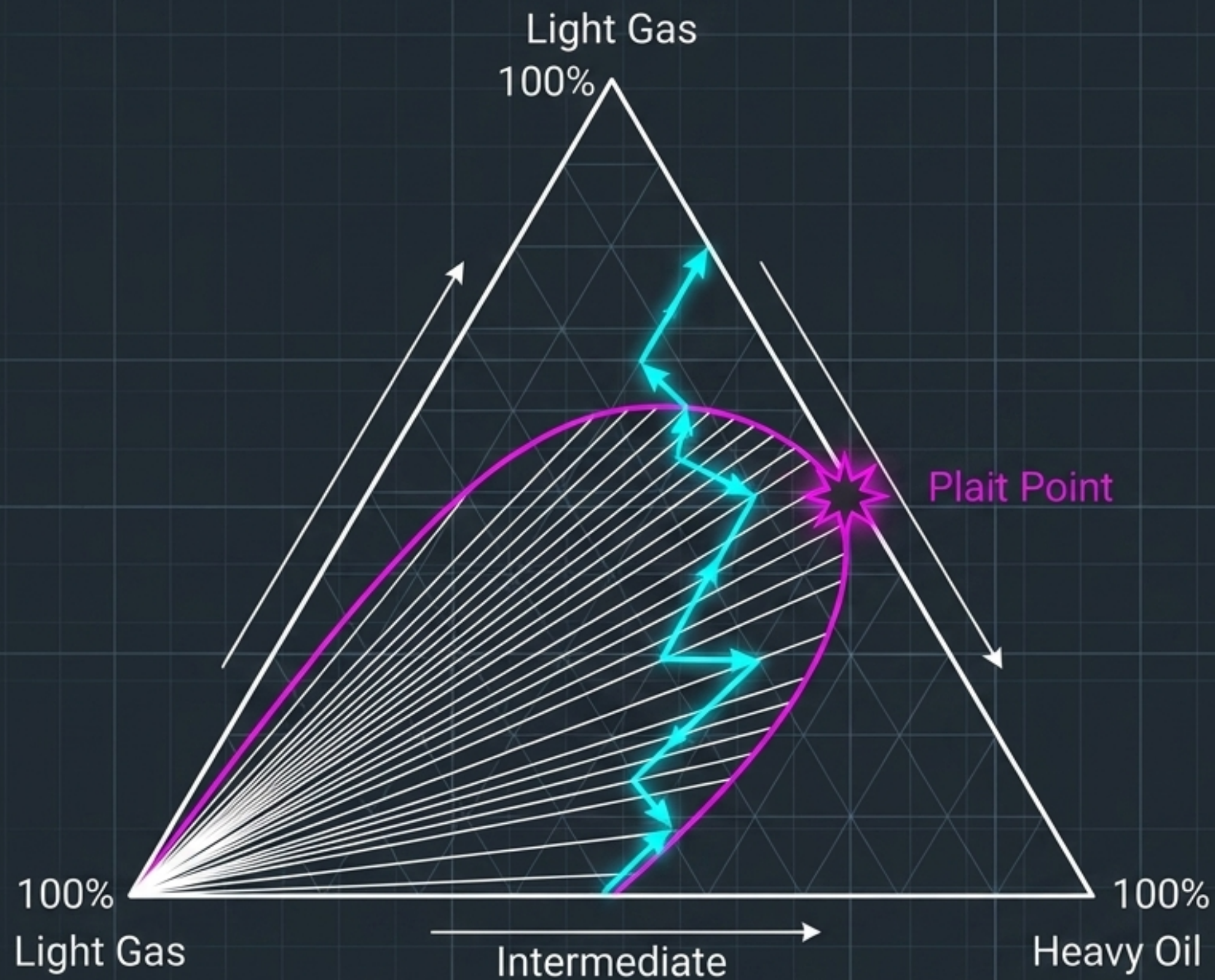
First-Contact Miscible (FCM)



Text Verdict

Instant, perfect dissolution upon contact. Maximum recovery.

The complete thermodynamic framework for predicting reservoir fluid behavior



Mastery Summary

1. Define pseudo-components (Coordinates)
2. Map phase stability (Binodal Curve)
3. Quantify volumetric splits (Lever Rule & Tie Lines)
4. Engineer recovery paths (Miscibility Design)

Thank You. Questions & Discussion.